BT1
3S LiPo 11.1V 6Ah
+12V_B
GND
SW1
Main switch
F1
15 A fuse
+12V
U1
Raspberry Pi 5
U2
ESP32-WROOM-32E
U4
INA219
U3
Mini560 5V/5A
U5
MD13S left
U6
MD13S right
J1
Motors JST
U7
Laser driver + nFET
D1
445 nm 2 W diode
SW2
Key lock
U8
HC-SR04 front-left
U10
VL53L0X front
LS1
Buzzer
U9
HC-SR04 front-right
U11
VL53L0X rear
D2
Power LED
U12
Servo yaw MG996R
U13
Servo shoulder DS3218
U14
Servo elbow DS3218
SW3
Emergency stop
U15
UBEC 6V 10A

Laser safety: Pi EN AND hardware key lock + e-stop kills motors+laser
Drive: 2x MD13S, 8 wheels; Sensors: 2x HC-SR04, 2x VL53L0X (I2C)
3S LiPo -> fuse/switch -> INA219 shunt -> buck 5V -> Pi 5 + ESP32

Semi-Autonomous Weeding Robot

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Title: Weed rover electronics

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